

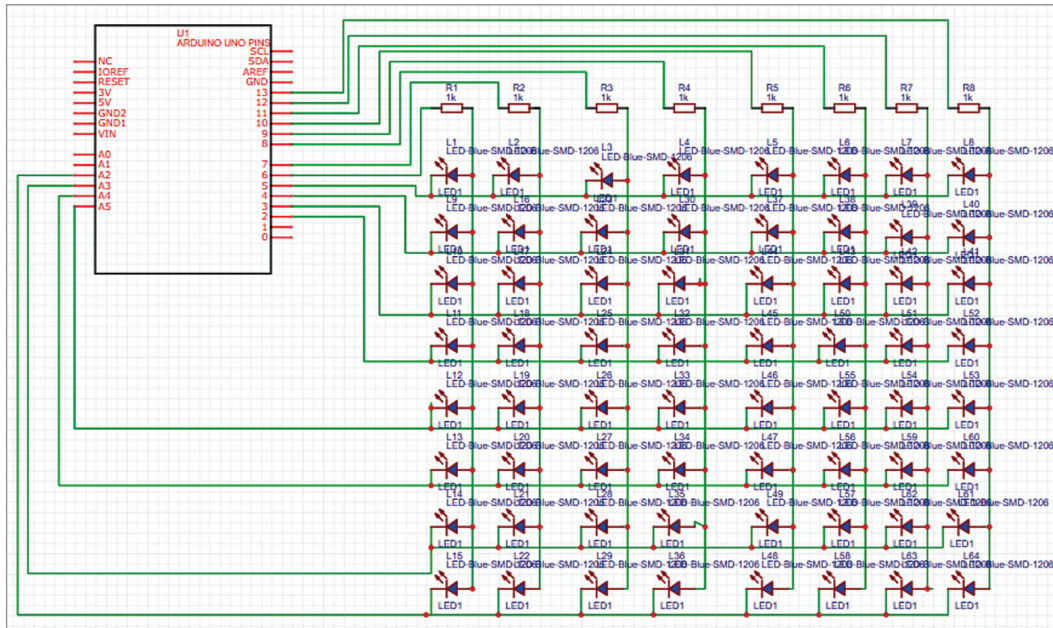


Fig. 3: Circuit diagram of the 8x8 SMD matrix display interfaced with Arduino Uno

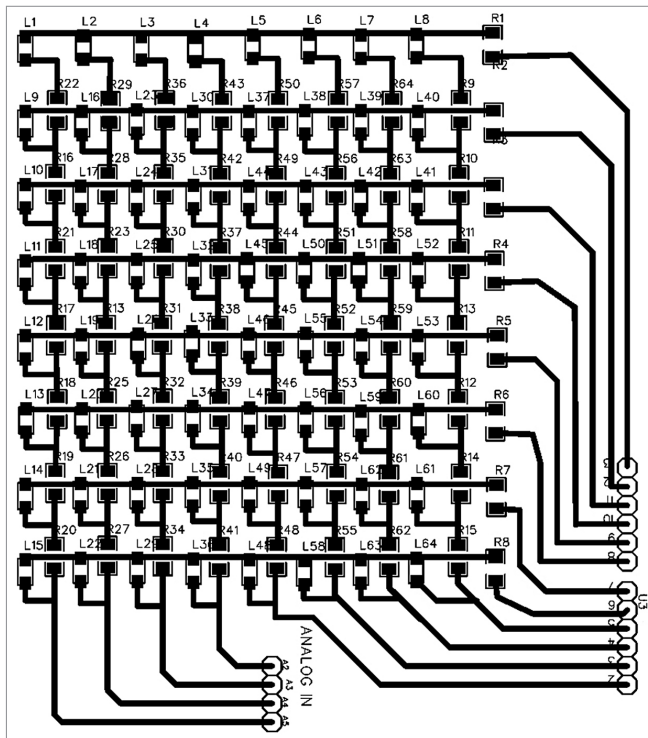


Fig. 4: Exported PNG file of the 8x8 SMD matrix display interfaced with Arduino Uno

If both the row and column of an LED are at the same level (either high or low), there is no potential difference across the LED, and it remains off. Thus, to control a particular LED, its row is set high and the corresponding column low.

selectively controlling the column states, multiple LEDs within a row can be operated simultaneously.

The system is programmed using the Arduino Uno, a widely used microcontroller board based on the ATmega328P, suitable for

controlling LED matrices and other embedded applications. The Arduino IDE is used to write, compile, and upload the code to the board.

While writing the code, pins are assigned for rows (R1-R8) and columns (C1-C8). In the setup() function, all row and column pins are configured as OUTPUT. The following functions are then defined:

- **on_row.**

Turns on a specific

row by setting the corresponding row pin high

- **on_column.** Turns on a specific column by setting the corresponding column pin low

- **off_row.** Turns off a specific row by setting the corresponding row pin low

- **off_column.** Turns off a specific column by setting the corresponding column pin high

- **off_all.** Turns off all rows and columns of the LED matrix by calling the offr and offc functions for



Fig. 5: Laser etching machine